

Scaffolding Low-Resource STEM Education with Large Language Models

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Abstract

Access to STEM higher education in low-resource language contexts is constrained by English-medium instruction and limited availability of localized technical resources. General-purpose machine translation tools (e.g., Google Translate) and static dictionaries fail to provide technical fidelity and conceptual explanations for specialized STEM terms. This study develops a cross-platform educational assistant powered by targeted Large Language Model (LLM) prompt engineering. The system translates English STEM terms into Burmese, generates concise Burmese conceptual explanations, and manages personalized saved glossaries for student review. This research contributes design principles for LLM-driven educational lexicography, demonstrating how structured prompt orchestration can bridge low-resource language barriers without costly model retraining.

Keywords: Low-Resource Languages, Large Language Models, STEM Terminology, Educational Technology, Design Science Research

1. Research Background

Access to higher education in Science, Technology, Engineering, and Mathematics (STEM) is heavily mediated through English-language literature and digital learning resources (Macaro et al., 2018). For non-native English speakers in developing regions such as Myanmar, this creates a severe cognitive burden, as students must simultaneously decipher complex technical concepts and foreign linguistic structures (Lin & Lei, 2021). Burmese, spoken natively by over 30 million people and by up to 10 million additional second-language speakers, remains a low-resource language with limited digital linguistic resources and language technologies, constraining the development of localized educational and STEM applications (Oo et al., 2020; Watkins, 2001).

To democratize higher education in under-resourced regions, Information Systems (IS) interventions must look beyond literal translation. Effective learning requires conceptual scaffolding—providing non-native speakers with accurate native-language terminology alongside clear contextual explanations (Walqui, 2006). However, low-resource languages often lack educational technologies capable of providing such support, creating barriers for students seeking access to global STEM knowledge.

2. Practical Problems

Given the scarcity of domain-specific Burmese digital language resources, students often resort to general-purpose tools such as dictionaries and machine translation systems (e.g., Google Translate), which are not designed to provide pedagogically appropriate explanations of technical concepts (Jolley & Maimone, n.d.; Oo et al., 2020). However, general-purpose machine translation systems often perform poorly for low-resource languages, particularly when translating specialized or domain-specific terminology, because of limited parallel corpora, vocabulary coverage, and training resources (Haddow et al., 2022a). Furthermore, static bilingual dictionaries provide only lexical equivalents and cannot keep

pace with rapidly evolving STEM terminology or offer the contextual, conceptual explanations needed for effective learning (Kasneci et al., 2023).

Without tailored explanations alongside translated terms, students may experience increased cognitive load and struggle to develop accurate conceptual understanding (Sweller, 1988). Furthermore, existing tools often lack personalized study features, such as reviewable word-translation pairs and retrieval-based practice, limiting students' ability to reinforce technical vocabulary through active recall (Dunlosky et al., 2013).

3. Research Problems or Gaps

While prior Information Systems (IS) and Natural Language Processing (NLP) literature has extensively examined machine translation and computer-assisted language learning, a critical gap remains in developing interactive terminology scaffolding systems for learners using low-resource scripts, where linguistic resources and educational technologies remain limited (Haddow et al., 2022b; Joshi et al., 2020). Extant machine translation research has primarily focused on sentence-level translation tasks and improving performance across language pairs, while domain-specific terminology management and contextualized explanations remain underexplored, particularly for low-resource (Haddow et al., 2022b).

Moreover, current IS research on Large Language Model (LLM) orchestration emphasizes enterprise Knowledge Management and Retrieval-Augmented Generation (RAG), leaving prompt engineering design patterns for low-resource educational dictionaries underexplored (Dwivedi et al., 2023; Lewis et al., 2021). Limited design knowledge exists regarding how lightweight API-driven LLM architectures can support accurate technical translation, contextual explanation, and interactive vocabulary management without resource-intensive model fine-tuning. This study addresses this gap by developing DSR-based design principles for LLM-powered educational assistants.

4. Research Objectives and Questions

To address these practical and theoretical gaps, this study designs, implements, and evaluates an interactive English-to-Burmese STEM learning assistant grounded in Design Science Research (Hevner et al., n.d.). The primary research objective is to formulate generalizable design principles for LLM-powered educational terminology assistants in low-resource script environments.

This study is guided by the following primary Research Question (RQ) and sub-questions:

- **Primary RQ:** *How can an LLM-powered educational Information System be designed to provide accurate STEM translations, conceptual explanations, and vocabulary scaffolding for students in low-resource language environments?*
 - **RQ1 (Design):** What prompt engineering patterns enable LLMs to generate technically accurate Burmese translations and accessible conceptual explanations for English STEM terms?
 - **RQ2 (Evaluation):** How do interactive terminology lookup, contextual explanations, and personal history features influence student usability and perceived learning support?

5. Research Methodology

This study adopts an adapted Design Science Research (DSR) methodology, using systems development as the primary research method to design and evaluate an LLM-driven educational dictionary artefact (Hevner et al., n.d.-a; Nunamaker & Chen, n.d.). The research follows four iterative phases.

- Phase 1: Problem Identification examines Burmese STEM terminology challenges and identifies gaps in low-resource language support and educational prompt engineering.
- Phase 2: Artefact Design and Development constructs few-shot LLM prompt templates and a cross-platform system for interactive terminology lookup and glossary management.
- Phase 3: Artefact Evaluation assesses translation quality through bilingual expert review and usability through the System Usability Scale.
- Phase 4: Reflection, Generalization and Communication derives Level 2 design principles for LLM-based educational dictionaries in low-resource contexts.

6. Artefacts Resulting from the Research

Following the DSR artifact typology (Hevner et al., n.d.), this research produces two primary artefacts.

- System Artefact (Software Application): a cross-platform web and mobile application that supports Burmese STEM terminology learning. The system accepts English technical terms, applies an LLM-based prompt pipeline to generate Burmese translations with concise conceptual explanations, and enables users to manage search histories and personalized vocabulary glossaries.
- Conceptual Artefact (Abstraction): a set of evaluated architectural patterns and prompt engineering principles representing Level 2 design knowledge (Hevner et al., n.d.). These principles describe how lightweight LLM services can be structured to support low-resource educational terminology translation and explanation generation without requiring model retraining.

7. Potential Structure of the Paper

This paper is structured as follows. Section 1 introduces the research context, practical motivation, and research gaps. Section 2 reviews relevant literature on low-resource language technologies, LLM-based educational support, and design science research. Section 3 presents the research methodology and DSR approach. Section 4 describes the design and development of the LLM-powered educational dictionary artefact. Section 5 presents the artefact evaluation and findings. Section 6 discusses the theoretical and practical contributions, including the resulting design principles, limitations, and future research directions. Section 7 concludes the paper.

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